



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Computer Networks lab Report - 12

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Setting up Dynamic Routing Protocol (RIP Version 1 & Version 2)

Objectives

The objectives of this lab are:

- To configure router hostnames.
- To set passwords for router security.
- To configure three Cisco routers.
- To implement dynamic routing using **RIP protocol**.
- To understand the difference between **RIP v1 and RIP v2**.

Theoretical Description

By default, Cisco routers use generic hostnames such as **Router**, which makes network management difficult. Assigning meaningful hostnames helps in identifying devices easily during configuration and troubleshooting.

Router security is implemented by setting:

- **Enable password**
- **Enable secret**
- **Console line password**
- **Auxiliary line password**

The **enable secret** password has higher priority than the enable password and is encrypted.

In large networks, **static routing** becomes complex and difficult to manage. Therefore, **dynamic routing protocols** are used to automate route learning.

Routing Information Protocol (RIP) is a distance-vector routing protocol that uses **hop count** as its routing metric. The maximum hop count supported by RIP is **15**, making it suitable for small networks.

RIP Version 2 improves upon RIP Version 1 by:

- Supporting **classless routing (VLSM)**
- Providing **authentication**
- Using **multicast updates** instead of broadcast

Lab Tasks and Configuration

This lab consists of multiple configuration tasks performed on **three routers**.

Task 1: Configure Hostname

Each router is assigned a unique hostname to improve identification.

Commands

```
Router(config)# hostname R1
```

The router name changes immediately after executing this command.

Task 2: Configure Enable Password

An enable password is set to restrict access to privileged EXEC mode.

Commands

```
R1(config)# enable password cisco
```

Task 3: Configure Enable Secret

Enable secret provides a more secure encrypted password.

Commands

```
R1(config)# enable secret cisco
```

Note: Enable secret overrides enable password.

Task 4: Configure Console Line Password

Console access is secured using a password.

Commands

```
R1(config)# line console 0
```

```
R1(config-line)# password lab
```

```
R1(config-line)# login
```

This ensures authentication when accessing the router through the console.

Task 5: Configure Auxiliary Line Password

Auxiliary line is secured for remote access.

Commands

```
R1(config)# line auxiliary 0
```

```
R1(config-line)# password ciscolab
```

```
R1(config-line)# login
```

Note: Auxiliary line password cannot be fully simulated in Packet Tracer.

Task 6: Configure IP Addresses on Router Interfaces

Each router interface is assigned an IP address according to the network topology.

Sample Commands

```
R1(config)# interface FastEthernet0/0
```

```
R1(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
R1(config-if)# no shutdown
```

The same steps are repeated for all routers and interfaces.

Task 7: Enable RIP Routing Protocol

RIP routing protocol is enabled on all routers.

Commands

```
R1(config)# router rip
```

Task 8: Configure RIP Version 2

RIP Version 2 is enabled to support classless routing.

Commands

```
R1(config-router)# version 2
```

Task 9: Advertise Networks

All directly connected networks are advertised using RIP.

Commands

```
R1(config-router)# network 192.168.1.0
```

```
R1(config-router)# network 192.168.2.0
```

Task 10: Disable Auto Summarization

Auto-summarization is disabled to allow proper subnet advertisement.

Commands

```
R1(config-router)# no auto-summary
```

Task 11: Verify RIP Routes

The routing table is checked to verify RIP-learned routes.

Commands

R1# show ip route

Routes learned via RIP are indicated with **R**.

Example:

R 192.168.2.0/24 [120/1] via 192.168.1.2

Result

- All routers were successfully configured with hostnames and passwords.
- RIP Version 2 was enabled on all routers.
- Routers dynamically learned routes using RIP protocol.
- Auto-summarization was disabled successfully.

Conclusion

In this lab, dynamic routing was implemented using RIP protocol. Router security was enhanced using passwords, and RIP Version 2 was configured to support modern networking features. The lab demonstrated how RIP simplifies routing in small-scale networks.